

King County House Price Prediction - AI Evaluation Report

Automated Machine Learning Benchmark and Explainable AI Assessment

Selected Champion Model

Model Architecture: XGBoost
R² Accuracy Score: 0.9067
Root Mean Squared Error (RMSE): \$92,210.07
Mean Absolute Error (MAE): \$61,230.36
Mean Absolute Percentage Error (MAPE): 12.42%

Model Benchmark Leaderboard

Model	MAE (\$)	RMSE (\$)	R ² Score	Adj R ²
XGBoost	\$61,230	\$92,210	0.9067	0.9062
CatBoost	\$62,622	\$93,613	0.9039	0.9033
Gradient Boosting	\$63,600	\$95,687	0.8996	0.8990
LightGBM	\$64,913	\$97,408	0.8959	0.8953
Random Forest	\$64,123	\$99,732	0.8909	0.8903
Extra Trees	\$63,936	\$100,008	0.8903	0.8897
Decision Tree	\$84,843	\$135,103	0.7998	0.7986
Linear Regression	\$103,974	\$145,786	0.7669	0.7655
Ridge Regression	\$104,057	\$146,043	0.7661	0.7647
Lasso Regression	\$104,127	\$146,229	0.7655	0.7641
AdaBoost	\$121,054	\$165,816	0.6984	0.6967